Constraint Torques at the Wrist: Grip Geometry, Energy Transfer and Face Control

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Contents

This print companion preserves the original manuscript's author credit and follows the corrected AffineDrift web article. Its idealized derivations support a testable hypothesis; they do not establish a human grip ranking.

Abstract

Grip geometry can change how the golfer's motion and forces affect the clubface. Understanding that effect requires more than resolving a torque into two axes. The wrist's allowed motion determines its reaction loads; the hand–club attachment transforms those loads; the club's inertia and moving support determine its response; and the face-orientation map determines which response matters at impact. Muscle action, compliance and correlations between disturbances enter the same chain. This article derives those connections using an ideal two-axis wrist model, then identifies the measurements needed to qualify it. The result is a testable grip-geometry hypothesis, not a demonstrated ranking of finger and palm grips. Mechanical identities and illustrative numerical examples are verified here; human performance and the author's historical Simscape model are not validated by those checks.

In Plain Language

A joint can pass a load between body segments without behaving like an extra motor. Changing how the club sits in the hand can change the motion that this load produces. But a load that is less disruptive in one configuration may become larger there, and the golfer may change muscle action at the same time. The useful question is therefore: **which measured grip geometry produces the desired club delivery, with manageable effort and sensitivity, for this golfer?** The equations below make each part of that question explicit.

Introduction

The investigation began with a concrete simulation problem. A model was driven with prescribed joint motion, and the joint reported forces and torques. Could those recorded values be fed back as actuator inputs to reproduce the motion in forward dynamics? The obstacle was that a measured joint wrench and a set of actuator inputs are different quantities. A joint can report three torque components while having only two rotational coordinates. Some of its load enforces the joint geometry rather than driving an independent rotation.

That observation motivates a broader golf question. Does the hand–club attachment direct these loads into motion that develops useful clubhead speed, or into motion that makes face control more sensitive? This is a worthwhile mechanical hypothesis. Its answer depends jointly on geometry, dynamics and the task. We will retain the question while replacing claims of automatic transmission efficiency, inevitable variability accumulation and universal grip superiority with quantities that can actually be calculated and tested.

The Wrist as a Universal Joint

We distinguish the **wrist complex** from the **forearm's radioulnar joints**. Flexion–extension and radial–ulnar deviation describe two prominent directions of hand motion relative to the forearm. Pronation–supination involves the radius and ulna and changes the frame carried to the wrist. It should be modeled at the appropriate forearm joints, not counted as an unactuated third wrist coordinate after it has already been prohibited by a two-coordinate model.

The ideal wrist used here consists of two intersecting revolute axes. Near a chosen neutral posture, associate flexion–extension with a mediolateral axis and deviation with an approximately dorsopalmar axis. Declare the anatomical calibration and signs for the particular limb. These are modeling coordinates, not a claim that anatomical or mechanical axes remain perfectly perpendicular and fixed throughout a swing. Crisco and colleagues found oblique mechanical axes in a six-cadaver-wrist study; their separate in-vivo study examined carpal kinematics during dart-throwing motions. Neither study validates the present golf model. Crisco et al., 2011, Crisco et al., 2005.

Limitations of the 2-DOF Wrist Model

A rigid joint is an approximation to motion under specified loads and time scales. It can be useful when omitted relative motions have negligible influence on the output being studied. That must be checked, especially for tissue loads, contact transients and impact. A compliant extension introduces extra states, constitutive parameters and possibly activation-dependent stiffness. Calling an approximation "rigid" neither validates it nor makes it inherently invalid.

Two coordinates with two independent generalized torque inputs describe a locally fully actuated wrist subsystem. Underactuation must instead be assessed on the independent coordinates of the complete model, including its floating base, contacts and actuator map. This is also a **spatial** model: a two-axis joint is not a planar swing simply because it has two rotational coordinates. Planar tests elsewhere in the project cannot validate its out-of-plane face-orientation predictions.

Universal Joints and Constraint Torques

Mechanical Properties of Universal Joints

The perpendicular axes in a universal joint belong to its cross pins. The centerlines of the connected shafts can form a range of bend angles; they are not required to be perpendicular. Furthermore, a freely moving two-axis relative joint and a driveshaft held in two fixed bearings are different mechanisms. The bearings impose additional constraints. The familiar Cardan speed-ratio formula describes the latter assembly and is derived separately below. It cannot be assigned directly to wrist flexion and grip angle.

Let $R = {}^F R_H = R_x(\phi)R_y(\psi)$ map hand-frame vectors into the forearm frame. At the joint's common point, the two allowed relative angular velocity directions, expressed in the forearm frame, are

$$\mathbf{a} = \mathbf{e}_x, \qquad \mathbf{b} = R_x(\phi)\mathbf{e}_y, \qquad \omega_{H/F} = \mathbf{a}\dot{\phi} + \mathbf{b}\dot{\psi}.$$

The ideal reaction couple must do zero power against every allowed relative angular velocity. Thus it lies along the reciprocal direction

$$\mathbf{n}_c = \mathbf{a} \times \mathbf{b} = R_x(\phi)\mathbf{e}_z, \qquad \tau_c = \lambda_r \mathbf{n}_c, \qquad \mathbf{n}_c^T \omega_{H/F} = 0.$$

For this convention a local holonomic orientation constraint is $\mathbf{e}_x^T R \mathbf{e}_y = 0$. Its derivative gives the same instantaneous restriction. The joint also imposes three translational constraints at the common point, whose reactions are forces. If a wrench is reported at a different point, those forces contribute an additional moment.

Derivation for the Wrist Universal Joint

At $\phi = 30^\circ$, $\psi = 20^\circ$, the reciprocal unit direction is

$${}^F\mathbf{n}_c = (0, -0.5, 0.866025)^T, \qquad {}^H\mathbf{n}_c = R^T\,{}^F\mathbf{n}_c = (-0.342020, 0, 0.939693)^T.$$

It is generally neither a fixed forearm z-axis nor a fixed hand z-axis. The constraint does not say that a chosen Euler angle rate or the forearm-frame component ω_z vanishes. For example, the allowed deviation direction $\mathbf{b} = (0, \cos\phi, \sin\phi)^T$ has a nonzero z component when the wrist is flexed. Confusing coordinate rates with angular-velocity components would therefore label permitted motion "constrained."

Because $\mathbf{a}$ and $\mathbf{b}$ are orthogonal unit vectors, this joint's 3×2 angular-velocity Jacobian has rank two. Straight shaft alignment does not cause gimbal lock in this model. Singularities of a supported driveshaft assembly, an Euler chart, and a chosen golf task require their own analyses. They are not interchangeable.

The left panel shows the freely moving ideal wrist. The right panel includes fixed shaft supports and therefore describes a different mechanism. Its phase is not an anatomical wrist angle; the two panels must not be identified by matching their angle symbols.

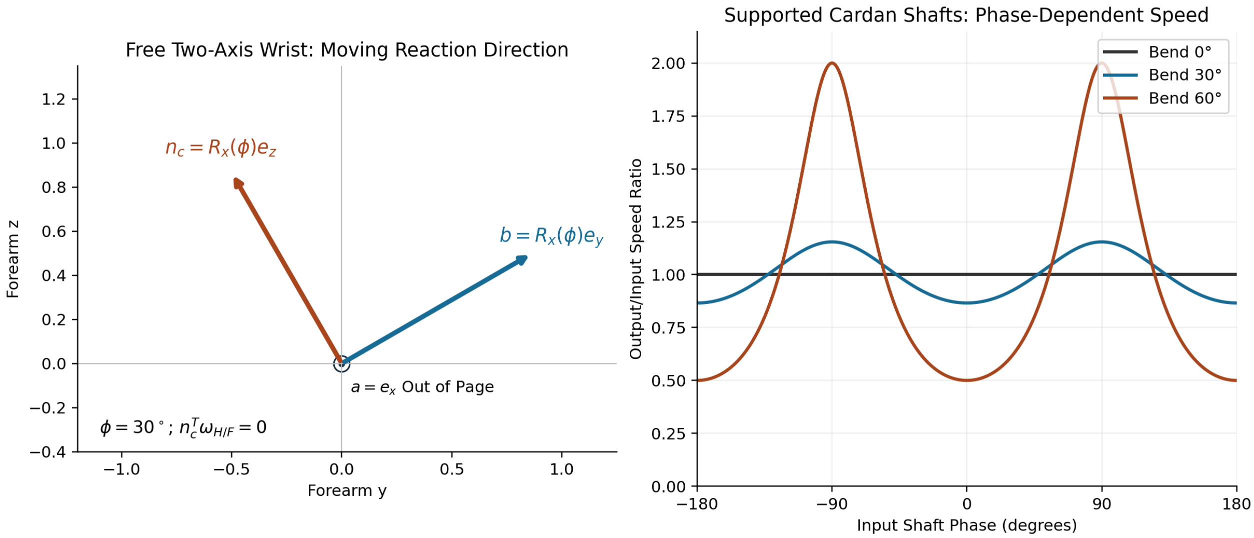

Figure 1: Two Different Mechanisms

Theoretical Formulation of Constraint Torques

Use redundant local generalized coordinates $\mathbf{q} \in \mathbb{R}^n$, an independent constraint Jacobian $A \in \mathbb{R}^{k\times n}$, and an actuator map $B \in \mathbb{R}^{n\times m}$. For stationary ideal constraints,

$$M\ddot{\mathbf{q}} + \mathbf{h} = B\mathbf{u} + \mathbf{f} + A^T\lambda, \qquad A\dot{\mathbf{q}} = 0.$$

M is positive definite in this coordinate chart. The bias $\mathbf{h}$ includes gravity and velocity-dependent inertial terms; $\mathbf{f}$ contains other declared generalized forces, including modeled passive forces if they are not placed in $\mathbf{h}$. Use one convention consistently. The components of λ are multipliers, not automatically anatomical torque components: their units and meaning depend on how the constraint equations are scaled. The physical generalized reaction is $A^T\lambda$.

Differentiate the velocity constraint and substitute the dynamics:

$$AM^{-1}A^T\lambda = -\dot{A}\dot{\mathbf{q}} - AM^{-1}(B\mathbf{u} + \mathbf{f} - \mathbf{h}).$$

With independent rows of A, the left matrix is positive definite. Solve this system rather than forming numerical inverses. In analytical notation,

$$\frac{\partial\lambda}{\partial\mathbf{u}} = -(AM^{-1}A^T)^{-1}AM^{-1}B.$$

A reaction can depend immediately on actuator input even though the joint has no actuator along its reciprocal direction. For a transparent local example, take $A = (0, 0, 1)$, $B = (\mathbf{e}_x, \mathbf{e}_y)$, zero bias, and the illustrative inertia matrix

$$M = \begin{bmatrix} 2 & 0 & 0.5 \\ 0 & 2 & 0 \\ 0.5 & 0 & 2 \end{bmatrix}.$$

An input $(u,0)^T$ gives $\ddot{q}_x = u/2$, $\ddot{q}_z = 0$ and $\lambda = u/4$. A changed actuator input changes the constraint torque at the same state. This counterexample is enough to disprove the blanket claim that all constraint loads belong to uncontrollable drift. It does not show that every desired reaction can be prescribed independently of motion.

Reactions annihilate allowed virtual velocities. They are not generally in the null space of B, which acts on input vectors, or Euclidean-orthogonal to every applied generalized force. Using a tangent basis Z with $AZ = 0$ eliminates ideal reactions from the reduced equations by multiplication with Z^T. To compute actual joint loads, recover them from the full balance. This is the distinction between reduced motion equations and reaction reconstruction. Modern Robotics, §8.7.

Constraint Torque Transmission in the Golf Swing

The Wrist as a Torque Transmission Pathway

At a common joint point P, let $(\mathbf{F}, \tau)$ be the ideal constraint wrench exerted on the hand by the forearm, expressed in one inertial frame. The opposing wrench acts on the forearm. Their combined power is

$$P_{c,H} + P_{c,F} = \mathbf{F}^T(\mathbf{v}_{P,H} - \mathbf{v}_{P,F}) + \tau^T(\omega_H - \omega_F) = 0.$$

The first term vanishes because the joint points coincide and have equal velocity; the second vanishes for the reciprocal couple. Yet the individual body powers need not be zero. For example, a $2\,\mathrm{N\,m}$ couple along $\mathbf{n}_c$ and a common $10\,\mathrm{rad/s}$ angular-velocity component along that direction give $+20\,\mathrm{W}$ to one body and $-20\,\mathrm{W}$ to the other. The ideal constraint redistributes energy without generating it.

This resolves the apparent contradiction between "does no work" and "transmits energy." Zero **combined** work is not zero work on each segment. If a proximal motion is externally prescribed, its driving actuator can supply energy to the modeled subsystem. For a moving constraint, the reduced subsystem reaction power need not vanish. Muscle work, tissue energy storage, dissipation, gravity and external contact power must be included in the appropriate system boundary. A large transmitted torque alone measures none of those efficiencies.

Inertia, Wrench Transport and a Moving Wrist

Describe grip attachment by a full rigid transform ${}^{H}T_C(g)$, where g collects measured grip parameters. The attachment rotation and translation, along with any separate compliance model, cannot generally be reconstructed from a single "finger versus palm" label. For a force applied at P, a moment reported about club center of mass C is

$$\tau_C = \tau_P + (\mathbf{p}_P - \mathbf{p}_C) \times \mathbf{F}.$$

Then rotate both force and moment into the desired frame. A pure couple is independent of reference point, but a wrench with nonzero force is not. Similarly, express and transport the full club inertia tensor:

$$I_P = I_C + m\big[(\mathbf{r}^T\mathbf{r})\,\mathrm{Id} - \mathbf{r}\mathbf{r}^T\big], \qquad \mathbf{r} = \mathbf{p}_C - \mathbf{p}_P.$$

At the center of mass, Euler's equation is $\sum \tau_C = I_C\dot{\omega} + \omega \times (I_C\omega)$. For a body-fixed point P accelerating with $\mathbf{a}_P$, the corresponding balance is

$$\sum \tau_P = I_P\dot{\omega} + \omega \times (I_P\omega) + m\mathbf{r} \times \mathbf{a}_P.$$

All vectors in an equation must share a frame. The last term is why dividing a wrist moment by a grip-point inertia is not a complete moving-club model. Off-diagonal inertia and velocity coupling can also matter. For a fixed pivot, zero instantaneous angular velocity and a free rotational response, $\dot{\omega} = I_P^{-1}\tau_P$. A one-axis constrained response instead uses that axis's scalar inertia and supporting reactions.

For scale, a point head of mass 0.200 kg at 0.85 m from the grip and a thin uniform 0.100 kg shaft of length 1.0 m give a **transverse** grip-point inertia

$$I_\perp = 0.200(0.85)^2 + \tfrac{1}{3}(0.100)(1.0)^2 = 0.177833\,\mathrm{kg\,m^2}.$$

This idealized geometry does not determine a realistic shaft-axis inertia. That needs shaft radius, the head's intrinsic tensor and its offset from the shaft. The earlier $0.005\,\mathrm{kg\,m^2}$ value cannot be called the whole club's transverse grip inertia for this geometry. An assumed inertia ratio of 2, 0.5 or 30 does not become a measured club property by appearing in a simulator.

The Grip Angle Hypothesis

Coordinate System Definitions

Define a ground frame with x toward the target, z upward and y completing a right-handed frame. Let $\mathbf{s}$ denote the unit shaft direction and $\mathbf{n}_f$ the outward face normal. A swing plane, when used, needs an explicit construction and time interval. Rotation within that plane is about its **normal**, not an axis lying in it. Shaft rotation is about $\mathbf{s}$. Neither axis is automatically a principal inertia axis or a pure face-yaw axis.

Define face yaw by the horizontal projection of its normal:

$$\gamma = \operatorname{atan2}(n_{f,y}, n_{f,x}), \qquad n_{f,x}^2 + n_{f,y}^2 > 0.$$

For a small spatial orientation perturbation $\delta\theta$, $\delta\mathbf{n}_f = \delta\theta \times \mathbf{n}_f$, so

$$\delta\gamma = \frac{n_{f,x}\delta n_{f,y} - n_{f,y}\delta n_{f,x}}{n_{f,x}^2 + n_{f,y}^2} = H_\gamma\,\delta\theta.$$

At a square face with loft ℓ, choose $\mathbf{n}_f = (\cos\ell, 0, \sin\ell)^T$. Then $H_\gamma = (-\tan\ell, 0, 1)$. Rotation about the vertical changes yaw; rotation about the target direction also changes yaw when loft is nonzero. Rotation about the face normal itself does not change that normal. This derivation replaces an undefined "shaft angle" multiplier. A local derivative also becomes poorly conditioned when the normal is nearly vertical; changing the output definition may then be necessary.

Grip Position Is a Geometric Parameter, Not a Verdict

In the ideal attachment, $R_{WC} = R_{WF}R_{FH}R_{HC}(g)$. Grip changes can alter the expressed reaction direction, the moment arm of transmitted forces, the effective inertia and the face measurement map. In a complete model they can also alter allowable wrist posture, passive impedance, muscle moment arms and contact closure through the second hand. A torque projection holds only the first of these effects in view.

Under an intentionally restricted short-time calculation—fixed pivot, frozen geometry, zero initial angular velocity and no other torque change— a perturbation of a reaction couple gives

$$\delta\ddot{\gamma} \simeq H_\gamma I_P^{-1}\mathbf{n}_c\,\delta\lambda_r.$$

The scalar $H_\gamma I_P^{-1}\mathbf{n}_c$ is a **local sensitivity**, not a universal inertia ratio. A direction can have a small yaw sensitivity because of geometry, not simply because its inertia is large. The same direction may still change path, loft, strike location or speed. Moreover, λ_r itself changes with grip and inputs; comparing identical reaction noise between grips is an extra experimental assumption, not a consequence of the constraint equations.

Implications for Speed and Consistency

The hypothesis can now be stated precisely: some measured hand–club attachments may reduce the sensitivity of selected impact outputs to a specified perturbation distribution while preserving feasible delivery and effort. Another attachment may perform better under different disturbances, equipment or control policies. This is a design and identification problem.

Larger inertia reduces instantaneous acceleration for the same scalar torque in a specified one-axis model. It also changes the torque needed to achieve a desired motion. Inertia alone is not damping, does not dissipate energy, and does not guarantee less terminal angle error. Likewise, a smaller yaw sensitivity can coexist with greater speed variation. Study the joint output vector—speed, direction, face normal and strike—before calling a perturbation "performance-irrelevant."

Descriptions such as "in the fingers," "in the palm," "strong" or "weak" can help an instructor communicate, but they do not uniquely specify T_{HC}. They should be accompanied by measured geometry when testing this mechanism. No general optimum of 20–40 degrees, unavoidable palm-grip instability, or increase in clubhead speed follows from the derivation above.

Variability Accumulation Through the Kinematic Chain

State dependence and nonlinearity do not by themselves imply randomness, unpredictability or chaos. Let $\lambda = L(\mathbf{x}, \mathbf{u}, g, p)$, where p contains model parameters. A local perturbation has the form $\delta\lambda = L_x\delta\mathbf{x} + L_u\delta\mathbf{u} + L_g\delta g + L_p\delta p$. Its covariance includes all cross terms. Correlated variations can reinforce or cancel; a serial chain is not an arithmetic accumulator of independent errors.

Quantitative Example

To isolate integration, suppose a scalar torque error η is constant through a short interval T, has zero mean and standard deviation σ_τ, and acts on a single inertia I. Initial angle and velocity errors are zero. Direct integration gives

$$\delta\omega(T) = \frac{T}{I}\eta, \qquad \delta\theta(T) = \frac{T^2}{2I}\eta, \qquad \sigma_\theta = \frac{T^2}{2I}\sigma_\tau.$$

Using the earlier article's **illustrative**, unmeasured values $\sigma_\tau = 0.15\,\mathrm{N\,m}$ and $T = 0.010\,\mathrm{s}$:

Assumed Scalar Inertia	Acceleration SD	Velocity SD at T	Angle SD at T
$0.005\,\mathrm{kg\,m^2}$	$30\,\mathrm{rad/s^2}$	$0.300\,\mathrm{rad/s}$	$0.0015\,\mathrm{rad} = 0.08594^\circ$
$0.00015\,\mathrm{kg\,m^2}$	$1000\,\mathrm{rad/s^2}$	$10.000\,\mathrm{rad/s}$	$0.0500\,\mathrm{rad} = 2.86479^\circ$

These are rotations of the hypothetical scalar coordinate, not measured clubface yaw or shot dispersion. Their ratio is 33.333, and the integration factor is one-half. They establish a sensitivity under declared assumptions; they establish neither typical club inertias nor a grip comparison. Converting face orientation to ball flight would additionally require impact geometry, club velocity, ball/face interaction and aerodynamics.

Temporal Correlation and Feedback

For a time-varying scalar torque error with covariance $C_\tau(s,t) = \mathbb{E}[\eta(s)\eta(t)]$, the same linear integrator gives

$$\mathrm{Var}[\delta\theta(T)] = \frac{1}{I^2}\int_0^T\int_0^T (T-s)(T-t)C_\tau(s,t)\,ds\,dt.$$

Frozen error gives $\sigma_\tau^2 T^4/(4I^2)$. Ideal white torque noise with $C_\tau(s,t) = Q_\tau\delta(s-t)$ instead gives $Q_\tau T^3/(3I^2)$; Q_τ has torque-squared times time units and is not a pointwise torque variance. A sampled random trace also needs its sample interval, filtering and covariance specified. Resampling arbitrary independent values without those choices changes the physical disturbance model.

For a nonlinear swing linearized along a nominal trajectory, collect all perturbations in $\mathbf{z}$ and write $\delta\dot{\mathbf{x}} = A(t)\delta\mathbf{x} + E(t)\delta\mathbf{z}$. The transition matrix $\Phi(T,t)$ gives the terminal effect

$$\delta\mathbf{y}(T) = C_T\Phi(T,0)\delta\mathbf{x}(0) + \int_0^T C_T\Phi(T,t)E(t)\delta\mathbf{z}(t)\,dt.$$

Grip affects A, E and C_T, not just an isolated torque direction. Feedback changes the state evolution; feedforward preparation changes the nominal trajectory and correlations. Delays and actuator limits restrict compensation, but they do not justify labeling every reaction uncontrollable. If impact time itself changes, account for that event-time perturbation rather than sampling every trial at a fixed clock time and calling it "impact."

Connection to Arm-Club Plane Separation

Two vectors can lie in the same plane without being parallel. Consequently, "single plane" does not mathematically imply shaft–forearm collinearity, a palm grip, or alignment with the wrist's reciprocal direction. A plane defined by body landmarks and a plane fitted to a clubhead trajectory are also different constructions. Declare the definition before comparing them.

Coleman and Rankin's seven-golfer kinematic study found that the chosen left-arm plane changed during the downswing and the clubhead did not remain in it. That challenges a fixed-plane representation for those observations; it is not an experiment showing that deliberately creating plane separation improves grip transmission. Coleman and Rankin, 2005.

MacKenzie's later simulation examined how club position relative to the swing plane affects club delivery. Such a model can investigate a specific geometric mechanism under its imposed torques and objective, but does not establish a universal coaching rule or a causal finger-versus-palm effect. MacKenzie, club-position study.

Mathematical Framework: From Kinematics to Kinetics

Constraint Torques in Forward Dynamics Simulation

In Simscape Multibody, a primitive's actuator-torque output concerns its generalized actuation. Composite constraint torque and total torque are three-component vectors resolved in a selected base or follower frame. Total torque includes actuation, internal mechanics, limits and constraints. Changing "base on follower" to "follower on base" reverses the composite sign. These are software definitions; they do not identify which options were used in a historical model. MathWorks, Universal Joint, force and torque sensing.

Consequently, a difference between a two-input torque vector and a sensed three-component wrench is not evidence that mysterious extra torque has appeared. Their dimensions, bases and roles differ. Map the physical wrench to generalized forces by virtual power, and account separately for passive elements. Do not add "inertial torque" or "Coriolis torque" to a measured total wrench simply because those terms occur on the left of the equations of motion.

The Forward-Inverse Dynamics Challenge

For a prescribed, constraint-consistent trajectory, define $\mathbf{d} = M\ddot{\mathbf{q}} + \mathbf{h} - \mathbf{f}$. Inverse dynamics must solve

$$\begin{bmatrix} B & A^T \end{bmatrix} \begin{bmatrix} \mathbf{u} \\ \lambda \end{bmatrix} = \mathbf{d}.$$

The pair may be unique, underdetermined or infeasible, depending on actuation, contacts and the prescribed motion. Closed chains and two-hand grip loads can introduce internal-force indeterminacy. A least-squares answer with residual is not exact physical replay; report the residual and any force-allocation criterion. Reduced-coordinate inverse dynamics can determine net generalized joint torques without uniquely identifying muscle forces or internal contact loads.

To replay in forward dynamics, retain the same geometry, inertias, gravity, passive forces, external loads, contact mode and compatible initial conditions. Supply the recovered actuator inputs at their correct ports and allow reactions to be solved again. Remove prescribed-motion drivers that would otherwise overconstrain the replay. Compare motion, actuator work, joint loads and constraint residuals under tighter solver tolerances. The distinction is developed further in the inverse-dynamics article.

Physical Demonstration of Constraint Torque Generation

An instrumented mechanical jig can make this distinction tangible. Support a known two-axis joint, measure its configuration and applied primitive torques, and record the six-component load at a calibrated sensor. Compare the measured reaction with the constrained dynamics, including support motion and sensor moment arms. Use zero-load, quasi-static and dynamic trials to isolate terms.

Holding one's wrist with the opposite hand can suggest that loads are coupled, but it does not isolate a passive constraint torque. Voluntary co-activation, forearm action, external restraint and tissue deformation are all involved. The sensation is not an identification experiment or evidence that muscular effort about another axis is absent.

The Cardan Driveshaft Analogy

With shafts supported at a fixed bend δ, $|\delta| < \pi/2$, choose input phase φ and output phase χ so that

$$\chi = \operatorname{atan2}(\cos\delta\sin\varphi, \cos\varphi), \qquad \frac{\dot{\chi}}{\dot{\varphi}} = \frac{\cos\delta}{1 - \sin^2\delta\sin^2\varphi}.$$

Continue the angle through branch cuts for complete revolutions. The derivative has **no square root**. A phase origin shifted by 90° yields the cosine-squared form used in the Simscape Driveline documentation. Compare phase definitions before comparing extrema. MathWorks, Driveline Universal Joint.

For a lossless, negligibly inertial coupling with stationary supports, power-conjugate driving and delivered load magnitudes obey $\tau_{\text{out}}/\tau_{\text{in}} = \dot{\varphi}/\dot{\chi}$. With both port torques defined positive into the device, use the opposite sign in their power balance. Finite component inertia, compliance, losses or moving supports require their additional energy terms. Torque multiplication is not efficiency greater than one.

At $\delta = 30^\circ$, the speed ratio is 0.866025 at $\varphi = 0$ and 1.154701 at $\varphi = 90^\circ$; the ideal torque ratios are their reciprocals. Straight alignment gives a ratio of one everywhere. The limiting right-angle assembly is outside this operating model. Neither its degeneration nor an artificial 89-degree numerical clip represents an extreme anatomical palm grip.

The companion derivation separates this driveshaft calculation from the wrist hypothesis. A demo that uses its bend angle also as a torque-projection angle makes an additional synthetic choice. It does not solve wrist reactions, simulate the whole swing, or validate grip recommendations. Independent derivative and full-revolution checks are needed; multiplying two deliberately reciprocal formulas is only an internal consistency check.

Implications and Future Directions

Testable Predictions

Define a baseline participant, club, contact configuration, task and measured perturbation model. Then compare three questions separately:

Comparison	Held Fixed	Allowed to Change	What It Tests
Fixed-Input Forward Simulation	Actuator policy, initial state specification and external conditions	Motion and reactions under feasible new geometry	Response to a particular intervention
Fixed-Motion Inverse Analysis	A feasible target motion and external loads	Required actuation and internal-load allocation	Effort and load consequences of reproducing motion
Reoptimized Control	Objective, constraints and resource limits	Policy, trajectory and reactions	Best achievable result within the chosen model

Changing grip while imposing both all original torques and all original motion is generally not an independently specifiable comparison. Even initial states must be reconstructed to satisfy the new attachment and contact constraints. Report failures of feasibility rather than quietly relaxing the task for one condition. Speed and accuracy comparisons need the same definitions and units.

Simulation Studies

Start with an analytic joint whose reactions and power balance are known. Verify the spatial wrist kinematics, wrench transforms, inertia transport, constraint residuals and inverse-to-forward replay. Add the second hand and floating base with explicit closure, then add compliant structures when the question requires them. Test time-step, differentiation and parameter sensitivity before interpreting small differences between grips.

A useful result reports terminal-output covariance, actuator demand, tissue or contact loads within the model, and the perturbations responsible for each change. Check held-out perturbations and alternate plausible joint/club parameters. A result that reverses when a reasonable assumption changes is informative: it identifies what needs measuring before drawing a conclusion.

Experimental Validation

Measure the hand–club transform, wrist/forearm motion, club delivery and repeat-trial variability. Choose instrumentation and calibration appropriate to the predicted effect. Separate measurement error from between-trial motor variation, use participant-level comparisons, allow adaptation to grip changes, and report uncertainty rather than selecting only successful shots.

Motion capture and force plates help constrain a whole-body inverse problem, but they do not automatically identify how two hands share an internal grip load. Instrumented grip wrenches or justified contact/load-sharing models may be required. EMG measures electrical activity; it is not joint torque, muscle force or “control burden” without an additional measurement/model relationship. Use it as complementary evidence with its own normalization and uncertainty.

The hypothesis would be weakened if a predicted sensitivity change failed in held-out trials after measurement uncertainty and adaptation were accounted for. It would also be weakened if equally plausible contact or compliance models reversed the ranking. Finding a benefit for one participant or output would support that scoped result, not a universal rule for all golfers.

Extension to Other Joints

The same accounting applies to humanoid feet, elbows, the spine and robot grippers. A contact can remove motion directions while carrying forces; actuation changes both motion and reactions; compliant elements add storage and dissipation; and the chosen task decides which sensitivity matters. Changing contact mode can invalidate a local linearization. Contact force limits, friction and lift-off therefore belong in a whole-body analysis.

Theoretical Synthesis

Affine Decomposition of Constraint Forces

At a fixed state and contact mode, a torque-driven constrained model can be affine in its independent inputs. Eliminating reactions modifies both its input map and bias. Input-dependent muscle activation, stiffness, geometry or contact transitions may require a larger state or a different local model. "Not independently commanded" does not mean "independent of command."

Geometric Rejection of Disturbance

Geometry can reduce a chosen output's response to a particular load. To claim disturbance rejection over an interval, examine the complete propagation $C_T\Phi(T,t)E(t)$, not merely the instantaneous projection of a torque. A direction initially invisible to face yaw can affect it later through velocity, orientation or feedback coupling. This explains why the promising local idea needs a full dynamical treatment.

The Null Space of Performance

For a task map $\mathbf{y} = h(\mathbf{q})$, directions in $\ker(Dh)$ preserve the task to first order at that configuration. They need not preserve it after a finite step or preserve other delivery variables. The uncontrolled-manifold approach analyzes how observed variance relates to a specified task; it does not prove that the nervous system ignores those directions or that constraint noise automatically enters them. Scholz and Schöner, 1999.

Relationship to Instructional Practice

Grip advice may remain useful on the basis of practice, comfort, observation or other evidence. This article neither proves nor disproves that broader advice. It supplies a way to test one proposed mechanism. In practical terms, measure what changes with the grip, decide which delivery variables matter, and distinguish a geometrical explanation from evidence that an intervention improves a golfer's performance. A more complete model should sharpen that decision, not conceal assumptions behind more equations.

Conclusion

The wrist can transmit constraint loads while retaining two independent motion coordinates. Those loads are jointly determined by geometry, inertia, support motion, external forces and actuation. Their effect on golf delivery depends on the hand–club transform, the club's spatial dynamics, the face measurement map and the time structure of perturbations. The resulting grip hypothesis is conditional but constructive: identify geometries that make a feasible desired delivery less sensitive to specified disturbances, and test that prediction against measurements and alternative models. That connects the original simulation puzzle to a rigorous program for golf and humanoid mechanics.

Acknowledgments

The original discussion with Mike Duffey motivated questions about proximal motion and wrist reactions. Lynch and Park's Modern Robotics material provided the constrained-dynamics starting point. Discussion and analogy motivate the investigation; they are not substitutes for the derivations or validation.

Print and Interactive Companions

Download the Revised Print Companion. The Cardan and Torque Projection Demonstration illustrates the supported-shaft calculation and a synthetic torque projection. It does not calculate human wrist reactions or integrate clubface motion.

References

The annotated bibliography states which source supports which claim and distinguishes abstract-level review, technical documentation and empirical evidence. The main sources are linked at their points of use above. The companion validation guide defines reproducible mechanical checks and the remaining empirical questions.